

Research Statement

My research interest is the intersection of artificial intelligence, robotics, and the built environment. Specifically, I am interested in how robots can perceive, understand, and interact with complex construction environments to perform tasks autonomously and safely. This is because construction sites are among the most challenging environments for automation due to their dynamic, cluttered, and constantly changing nature, especially during low-light conditions. This motivates my research toward creating safer work surroundings by improving robot's understanding of construction environments to make reliable decisions and perform useful tasks under real-world conditions.

My interest in this topic has evolved gradually through my research experiences. Initially, during my master's studies, I worked on dimensional quality inspection of steel members using terrestrial laser scanned point cloud data. At the beginning, the goal was straightforward to obtain geometric information from point clouds and perform dimensional measurements. Later, I realized that scan data is often incomplete as steel member edge information was lost due to scan resolution limitations, which directly affected measurement accuracy. Instead of treating the missed information as unavoidable noise, I explored whether the geometric characteristics of steel members and scanner principles could be used to infer missing information. Thus, I developed an edge-loss compensation model that significantly improved the measurement accuracy. This experience taught me that successful automation depends not only on collecting data but also on understanding its limitations and extracting meaningful information from observations.

After graduation, I became interested in scene understanding of construction environments as a research assistant. I worked on semantic and instance segmentation of steel structure point clouds for Scan-to-BIM applications. Through this work, I explored both geometry-based and deep learning approaches for converting large-scale data into meaningful individual steel members. While deep learning methods demonstrated promising results, I observed that many construction applications suffer from limited training data and struggle when applied to highly specialized construction scenarios. I questioned myself that how domain knowledge, geometry, and learning-based approaches can be combined to improve perception in environments where data are limited.

To address this challenge, I began exploring alternative approaches such as simulation-based synthetic data generation and leveraging vision-language models and foundation models for training-free segmentation of indoor point clouds, investigating how pretrained semantic knowledge can be transferred to construction domains. From this, I deepened my understanding of how knowledge can be transferred across domains and how multimodal information from images and point clouds can be utilized more effectively.

While working on these perception-related directions, I also involved in construction robotics-related research, including studies on challenges in human-robot collaboration and challenges in robot application in construction sites. Through these experiences, I realized that perception alone is not sufficient for automation. A robot may accurately detect and classify objects, but it may still fail if it cannot understand the surrounding context, reason about the environment, and interact safely with people. This realization gradually shifted my interest from perception alone toward the broader challenge of enabling robots to understand and operate in complex construction environments.

Building on my experiences, I would like to pursue doctoral research on intelligent robotic systems that integrate perception, environmental understanding, and autonomous decision-making for human-robot collaboration in real-world complex construction applications. My long-term goal is to contribute to the development of practical robotic systems that can operate safely, efficiently, and reliably in real construction environments while addressing important challenges in the construction industry.